



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions



Dataset at a Glance

3,900

Total Purchases

Transactions analyzed

18

Data Points

Features per transaction

\$59.76

Avg Purchase

Per transaction

3.75

Avg Rating

Customer satisfaction

Comprehensive dataset covering demographics, purchase details, and shopping behavior across multiple categories



Data Preparation Journey

01

Data Loading & Exploration

Imported dataset, checked structure with pandas

02

Missing Data Handling

Imputed 37 missing review ratings using category medians

03

Feature Engineering

Created age groups and purchase frequency metrics

04

Database Integration

Loaded cleaned data into PostgreSQL for analysis

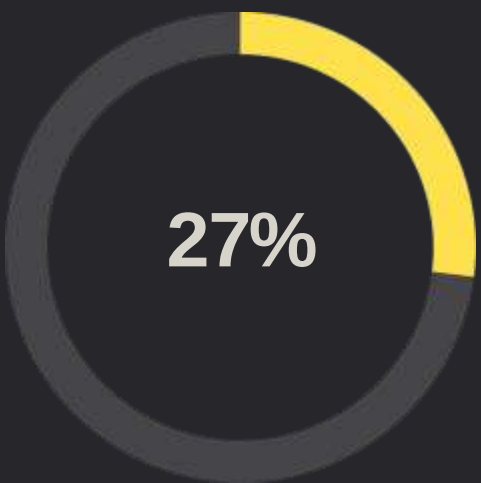
Revenue Insights: Gender & Subscriptions

Revenue by Gender

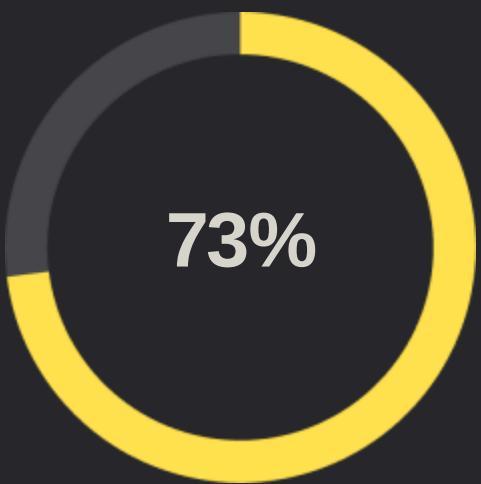


Male customers generate 68% of total revenue

Subscription Status



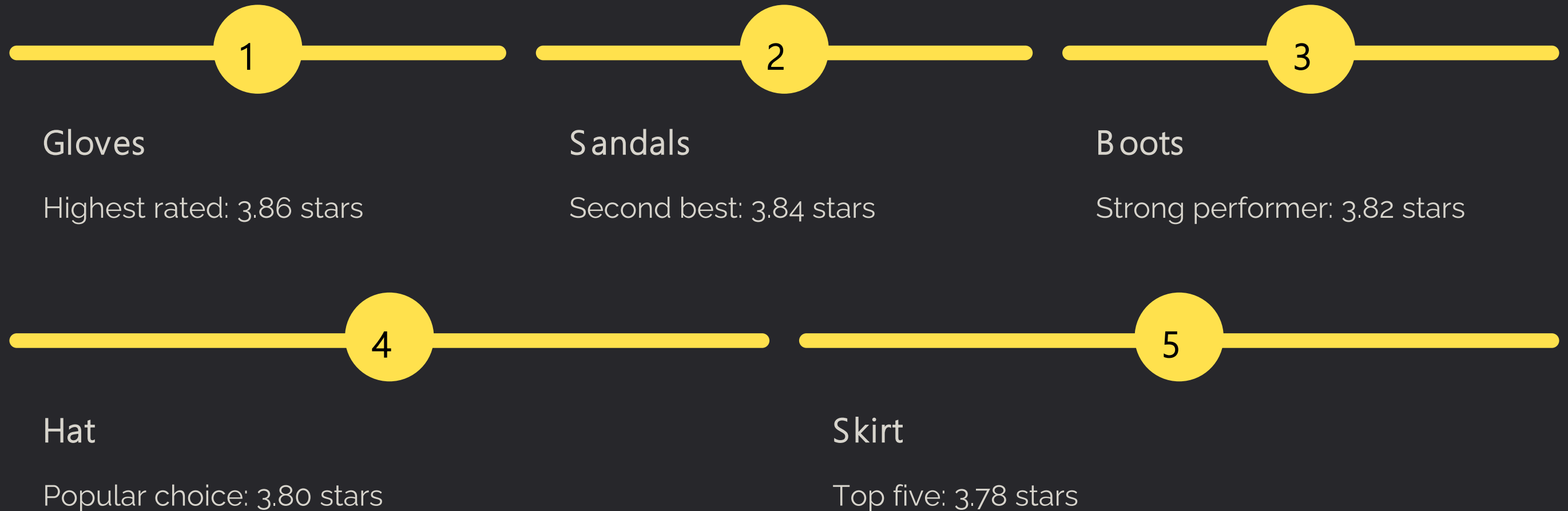
Subscribers



Non-subscribers

Avg spend: Subscribers \$59.49 vs Non-subscribers \$59.87

Top Performing Products



Discount Strategy Analysis

High-Spending Discount Users

839 customers used discounts but still spent above average

Opportunity to optimize discount targeting

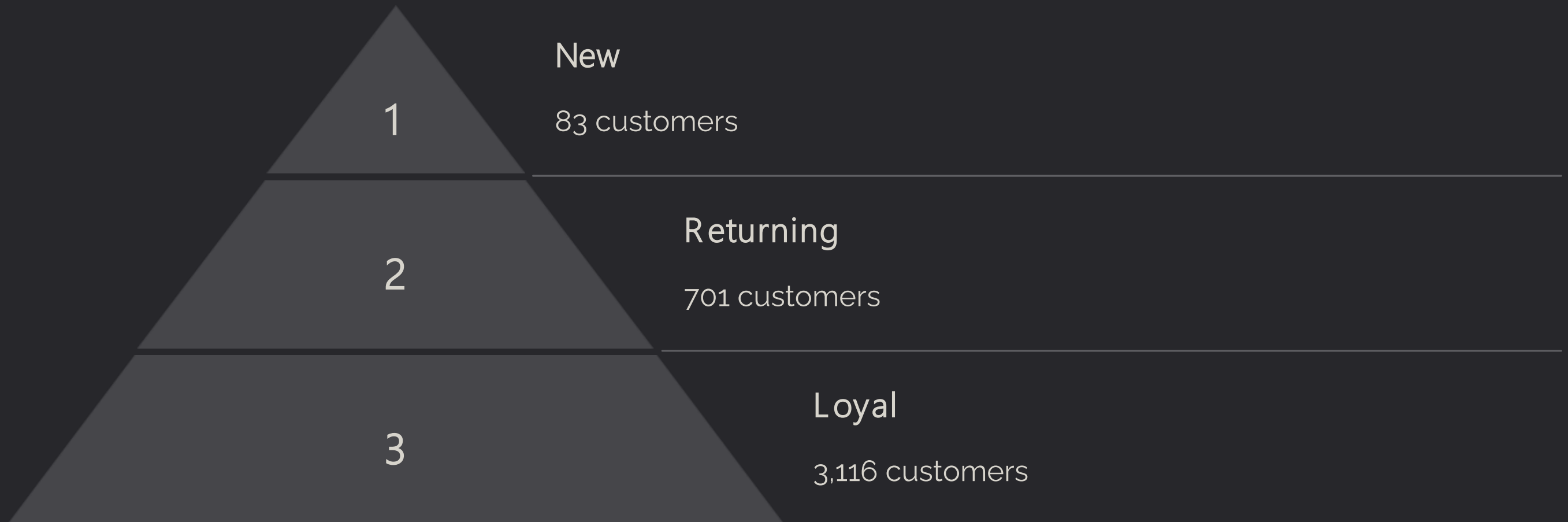
Most Discounted Products

1. Hat: 50% discount rate
2. Sneakers: 49.66%
3. Coat: 49.07%
4. Sweater: 48.17%
5. Pants: 47.37%



📌 Balance sales boosts with margin control—review discount policy for profitability

Customer Segmentation



80% of customers are loyal—focus on moving returning customers into this segment

Revenue by Age Group





Shipping & Category Performance

Shipping Type Impact

- Express: \$60.48 avg purchase
- Standard: \$58.46 avg purchase

Express shipping users spend 3.5% more

Top Categories

1. Clothing: Highest revenue & sales
2. Accessories: Strong second
3. Footwear: Consistent performer
4. Outerwear: Seasonal opportunity

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits to convert 73% non-subscribers



Loyalty Programs

Reward repeat buyers to grow loyal segment



Optimize Discounts

Balance promotions with margin control



Highlight Top Products

Feature high-rated items in campaigns



Targeted Marketing

Focus on young adults and express shipping users